

## PICKLE LAKE, ON, Canada

WMO: 718450

Lat: 51.4492N

Lon: 90.2181W

Elev: 386

StdP: 96.77

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |              |            |              | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|--------------|------------|--------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |              | 1%         |              |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB         | WS         | MCDB         | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-33.5 | (c)<br>-31.2 | (d)<br>-37.4                  | (e)<br>0.1 | (f)<br>-33.3 | (g)<br>-35.0 | (h)<br>0.1 | (i)<br>-30.9 | (j)<br>8.5            | (k)<br>-13.5 | (l)<br>7.9 | (m)<br>-14.0 | (n)<br>3.0            | (o)<br>280 | (p)<br>0.689 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.0                   | 28.5            | 18.8 | 26.6 | 17.4 | 25.0 | 16.7 | 20.7                | 26.0 | 19.3 | 23.8 | 18.2 | 22.6 | 4.6                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.9                            | 14.4 | 22.7 | 17.7 | 13.3 | 21.6 | 16.5 | 12.3 | 20.5 | 61.6          | 26.3 | 57.0 | 23.5 | 53.1 | 22.5 | 28.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 8.7           | (b) 7.7 | (c) 6.9 | (e) -36.8                  | (f) 32.3 | (g) 2.8            | (h) 2.3 | (i) -38.8                                          | (j) 33.9 | (k) -40.5  | (l) 35.2 | (m) -42.1  | (n) 36.5 | (o) -44.1  | (p) 38.2 |    |    |
| (a) 8.7           | (b) 7.7 | (c) 6.9 | (e) -36.9                  | (f) 22.8 | (g) 2.8            | (h) 1.8 | (i) -38.9                                          | (j) 24.1 | (k) -40.6  | (l) 25.1 | (m) -42.1  | (n) 26.1 | (o) -44.2  | (p) 27.3 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 0.8    | -18.8 | -15.9 | -8.1  | 0.3   | 8.4   | 15.3  | 18.1  | 16.8  | 11.3  | 3.1   | -6.3  | -15.2 |
|                                                                                  | DBStd       | 14.20  | 7.52  | 7.25  | 7.60  | 6.27  | 5.54  | 4.56  | 3.61  | 3.75  | 4.80  | 4.76  | 6.79  | 7.86  |
|                                                                                  | HDD10.0     | 4113   | 892   | 726   | 561   | 293   | 99    | 9     | 1     | 1     | 41    | 221   | 489   | 781   |
|                                                                                  | HDD18.3     | 6478   | 1151  | 959   | 818   | 540   | 310   | 111   | 48    | 76    | 216   | 471   | 739   | 1039  |
|                                                                                  | CDD10.0     | 769    | 0     | 0     | 1     | 3     | 50    | 168   | 251   | 211   | 79    | 7     | 0     | 0     |
|                                                                                  | CDD18.3     | 94     | 0     | 0     | 0     | 0     | 2     | 20    | 40    | 27    | 5     | 0     | 0     | 0     |
|                                                                                  | CDH23.3     | 774    | 0     | 0     | 0     | 1     | 28    | 193   | 317   | 201   | 32    | 1     | 0     | 0     |
|                                                                                  | CDH26.7     | 169    | 0     | 0     | 0     | 0     | 4     | 44    | 74    | 42    | 4     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 3.6    | 3.3   | 3.4   | 3.5   | 3.5   | 3.7   | 3.5   | 3.5   | 3.6   | 3.7   | 3.9   | 3.7   | 3.3   |
| Precipitation                                                                    | PrecAvg     | 735    | 32    | 28    | 35    | 41    | 72    | 93    | 104   | 88    | 86    | 64    | 54    | 38    |
|                                                                                  | PrecMax     | 921    | 59    | 45    | 75    | 108   | 118   | 188   | 166   | 159   | 203   | 102   | 115   | 72    |
|                                                                                  | PrecMin     | 533    | 15    | 12    | 13    | 10    | 21    | 29    | 34    | 27    | 11    | 22    | 24    | 14    |
|                                                                                  | PrecStd     | 115    | 11    | 9     | 14    | 23    | 29    | 45    | 35    | 38    | 45    | 27    | 23    | 15    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 0.7   | 3.8   | 12.5  | 19.4  | 26.2  | 30.7  | 32.0  | 30.4  | 26.6  | 19.6  | 10.5  | 2.3   |
|                                                                                  |             | MCWB   | 0.0   | 2.0   | 7.4   | 10.2  | 14.9  | 19.4  | 21.3  | 19.9  | 17.9  | 13.7  | 7.5   | 0.5   |
|                                                                                  | 2%          | DB     | -1.7  | 0.7   | 8.2   | 16.0  | 23.3  | 27.5  | 28.5  | 27.6  | 23.8  | 16.2  | 7.0   | 0.0   |
|                                                                                  |             | MCWB   | -2.2  | -0.8  | 4.0   | 7.9   | 13.2  | 17.1  | 19.1  | 19.0  | 17.0  | 11.6  | 4.6   | -0.7  |
|                                                                                  | 5%          | DB     | -3.9  | -1.5  | 5.3   | 12.7  | 20.9  | 25.5  | 26.6  | 25.6  | 21.3  | 13.2  | 4.5   | -1.8  |
|                                                                                  |             | MCWB   | -4.5  | -2.9  | 1.9   | 5.7   | 12.2  | 16.0  | 17.5  | 17.9  | 15.5  | 9.7   | 2.9   | -2.4  |
|                                                                                  | 10%         | DB     | -7.1  | -4.5  | 2.6   | 9.9   | 18.2  | 23.5  | 24.8  | 23.6  | 18.9  | 10.4  | 2.3   | -3.9  |
|                                                                                  |             | MCWB   | -7.6  | -5.6  | 0.0   | 4.1   | 11.1  | 14.9  | 16.9  | 16.8  | 14.3  | 8.1   | 1.2   | -4.4  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 0.1   | 2.3   | 7.8   | 11.2  | 17.7  | 21.6  | 22.4  | 21.9  | 20.0  | 14.9  | 8.3   | 0.9   |
|                                                                                  |             | MCDB   | 0.5   | 3.5   | 10.5  | 19.0  | 22.3  | 28.1  | 29.6  | 27.4  | 23.8  | 18.6  | 10.0  | 1.6   |
|                                                                                  | 2%          | WB     | -2.1  | -0.4  | 4.4   | 8.4   | 15.0  | 19.2  | 20.7  | 20.4  | 18.1  | 12.2  | 5.2   | -0.7  |
|                                                                                  |             | MCDB   | -1.6  | 0.6   | 7.4   | 14.4  | 20.1  | 24.2  | 25.9  | 25.5  | 21.8  | 15.3  | 6.5   | 0.0   |
|                                                                                  | 5%          | WB     | -4.5  | -2.8  | 2.1   | 6.3   | 13.1  | 17.6  | 19.3  | 19.1  | 16.5  | 10.1  | 3.1   | -2.4  |
|                                                                                  |             | MCDB   | -4.0  | -1.6  | 4.9   | 11.7  | 18.8  | 22.3  | 23.9  | 23.1  | 20.4  | 12.5  | 4.2   | -1.8  |
|                                                                                  | 10%         | WB     | -7.6  | -5.6  | 0.3   | 4.5   | 11.5  | 16.2  | 18.1  | 17.9  | 14.9  | 8.1   | 1.1   | -4.4  |
|                                                                                  |             | MCDB   | -7.2  | -4.3  | 2.2   | 9.6   | 17.3  | 21.4  | 22.6  | 22.3  | 18.1  | 10.3  | 2.2   | -3.9  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.8   | 10.7  | 11.6  | 11.4  | 11.1  | 10.6  | 10.0  | 9.6   | 8.6   | 6.4   | 6.3   | 7.8   |
|                                                                                  |             | MCDBR  | 9.6   | 9.8   | 12.1  | 14.9  | 14.6  | 13.3  | 12.3  | 12.2  | 12.1  | 11.1  | 7.5   | 7.7   |
|                                                                                  |             | MCWBR  | 9.2   | 8.5   | 8.4   | 8.8   | 7.1   | 5.9   | 5.4   | 5.6   | 6.6   | 7.4   | 5.8   | 7.1   |
|                                                                                  | 5% WB       | MCDBR  | 9.3   | 8.8   | 11.3  | 13.8  | 12.5  | 10.8  | 10.8  | 10.5  | 10.2  | 10.4  | 7.2   | 7.6   |
|                                                                                  |             | MCWBR  | 9.1   | 8.0   | 8.3   | 8.9   | 7.0   | 5.9   | 5.8   | 5.8   | 6.8   | 7.4   | 5.9   | 7.1   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.241  | 0.262 | 0.273 | 0.314 | 0.337 | 0.348 | 0.373 | 0.349 | 0.330 | 0.288 | 0.280 | 0.233 |       |
|                                                                                  | taud        | 2.319  | 2.282 | 2.365 | 2.437 | 2.484 | 2.479 | 2.433 | 2.510 | 2.557 | 2.609 | 2.440 | 2.373 |       |
|                                                                                  | Ebn at Noon | 801    | 874   | 937   | 929   | 918   | 907   | 875   | 883   | 866   | 851   | 745   | 759   |       |
|                                                                                  | Edh at Noon | 61     | 85    | 96    | 102   | 102   | 104   | 107   | 94    | 80    | 63    | 56    | 49    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.15   | 2.40  | 3.74  | 5.08  | 5.17  | 5.45  | 5.38  | 4.48  | 3.16  | 1.84  | 0.95  | 0.71  |       |
|                                                                                  | RadStd      | 0.13   | 0.19  | 0.34  | 0.40  | 0.36  | 0.47  | 0.51  | 0.41  | 0.36  | 0.25  | 0.10  | 0.12  |       |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (3 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page